FROST II: certified Hartree–Fock limits of the closed-shell atoms from neon to xenon

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We extend FROST (Fock radial orbital shooting with Taylor series) from atoms whose occupied orbitals are all s orbitals to closed-shell atoms with p and d shells, and report rigorous upper bounds on the Hartree–Fock (HF) energies of the closed-shell atoms from neon to xenon. Their leading digits are correct to 504 significant figures for neon, 103 for magnesium, 105 for argon and 103–106 for the heavier atoms, against previous records of 30 and 25. Three changes to the method make this possible. First, exchange with multipoles $k \geq 1$ gives every canonical orbital the exponential tail of the outermost one, so all orbitals must share one wall; this also makes them exactly orthogonal. Second, over that common wall the inner shells are extremely stiff: the growing mode of the krypton $1s$ orbital rises by 10^{1701}. The equations are therefore solved by multiple shooting, with Taylor-series segments short enough that no mode grows by more than 10^g, coupled by continuity conditions and solved by Newton's method with a sparse LU factorization, so that the working precision grows with the target accuracy rather than with the stiffness. Third, a homotopy in the electron–electron coupling, from hydrogenic orbitals, supplies a starting point that converges to the ground state. The exact-rational certificate carries over to any angular momentum. Writing each radial function as $P = r^{l+1}f$ with f piecewise polynomial, the triangle rule $k \leq l_a + l_b$ cancels the factor r^{-k-1} of every multipole, so all overlap, one-electron and Slater integrals are exact rationals. All 25 published digits of the quadruple-precision energies of Cinal are reproduced, as are his positions of the exchange-induced nodes of inner orbitals, which we locate to about 100 significant figures.

I. INTRODUCTION

In a previous paper [1] (paper I) we obtained the Hartree–Fock (HF) limits of helium and beryllium to 503 and 501 significant figures, without a basis set. The radial HF equations were integrated by arbitrary-order Taylor series in arbitrary-precision arithmetic, the orbital energies were fixed by shooting to a hard wall, and an independent program turned the resulting piecewise-polynomial orbitals into an exact rational upper bound on the energy. Paper I treated only atoms whose occupied orbitals are all s orbitals. Most closed-shell atoms are not of this kind.

The HF limits of heavier atoms have been computed by many numerical methods: finite differences [2–4], B-splines [5], finite elements [6] and Slater-type expansions [7]. In double precision these give 10–14 significant figures. Cinal's pseudospectral method in quadruple precision reached 25 figures for all 18 closed-(sub)shell atoms from helium to nobelium [8], and Yamamoto and Hatano have recently reported 30-figure energies for atoms up to the third period with Laguerre-type Λ functions, also in quadruple precision [9]. No HF energy of a closed-shell atom beyond beryllium has been known to more than 30 figures.

Three things change when the occupied orbitals include p and d shells.

- **Multipoles.** Exchange between orbitals of different angular momentum involves the multipoles $r_<^k/r_>^{k+1}$ with $k \geq 1$. Each needs its own pair potential, and krypton already has 47 of them.

- **Tails.** A nonvanishing multipole moment couples an inner orbital to the tail of the outermost one. As Handy, Marron and Silverstone showed [10], every canonical HF orbital then decays with the exponent of the highest occupied orbital. In paper I each beryllium orbital had its own wall; with p and d shells that cuts off the inner orbitals' tails, and one common wall is required (Sec. II B).

- **Stiffness.** With a common wall, the inner orbitals' growing modes must be controlled over the whole range. For krypton, $e^{\kappa_{1s}R}$ with $\kappa_{1s} = 32.25$ and $R = 121.44$ bohr is 10^{1701}, far beyond any working precision. Paper I's nested single shooting cannot cope with this.

The contributions of this paper are:

1. A formulation in which every orbital and every multipole potential obeys an ODE of the same form, $ru'' + 2(L+1)u' = g$, so that a single Frobenius–Taylor recurrence serves all of them, with one common wall (Sec. II).

2. Multiple shooting with Taylor segments, a homotopy in the electron–electron coupling, and wall continuation. Together they give the HF limit to any number of digits with working precision of about the target plus a stiffness margin that does not depend on κ_{1s}/κ (Sec. III).

3. The exact certificate for any angular momentum (Sec. IV).

4. Certified HF limits for Ne, Mg, Ar, Ca, Zn, Kr, Sr, Pd, Cd and Xe, which agree with all published digits [8], and the exchange-induced nodes of their inner orbitals to about 100 digits (Sec. V).

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II. FORMULATION

This section puts the HF equations into a form that a single numerical tool can handle: we first choose variables in which every equation has the same shape (Sec. II A), then show why all orbitals must share one wall (Sec. II B), and finally write the energy so that it can later be evaluated exactly (Sec. II C).

A. Equations

Each orbital a, with quantum numbers (n_a, l_a), is doubly occupied in all $2l_a+1$ magnetic sublevels, and its normalised radial function $P_a(r)$ behaves as r^{l_a+1} near the nucleus. Dividing out this factor, we work with

$$f_a = P_a/r^{l_a+1}, \tag{1}$$

which is an ordinary power series at $r=0$. The electrons interact through the multipole potentials

$$Y^k_{ab}(r) = r\int_0^\infty P_a(s)P_b(s)\frac{r^k_<}{r^{k+1}_>}\,ds, \tag{2}$$

one for each pair of orbitals and each $k = |l_a-l_b|, |l_a-l_b|+2, \ldots, l_a+l_b$, and since these behave as r^{k+1} near the nucleus we likewise work with $y^k_{ab} = Y^k_{ab}/r^{k+1}$. In these variables the canonical closed-shell HF equations are

$$\begin{aligned} r f_a'' + 2(l_a+1)f_a' = 2\Big[&\Big(r\Big(2\sum_b (2l_b+1)y^0_{bb} - y^0_{aa}\Big) \\ &- Z - \varepsilon_a r\Big) f_a \\ &- \sum_{(b,k)\neq(a,0)} e^k_{ab}\, r^{k+1+l_b-l_a}\, y^k_{ab} f_b\Big], \end{aligned} \tag{3}$$

$$r y^{k\,\prime\prime}_{ab} + 2(k+1)y^{k\,\prime}_{ab} = -(2k+1)\, r^{l_a+l_b+1-k} f_a f_b, \tag{4}$$

with $e^k_{ab} = (2l_b+1)\begin{pmatrix} l_a & k & l_b \\ 0 & 0 & 0\end{pmatrix}^2$. The first term of Eq. (3) is the Coulomb potential of all the electrons less the $k=0$ self-exchange of orbital a, the next two are the nucleus and the orbital energy, and the last sum is the remaining exchange. Equation (4) is Poisson's equation for each multipole. Because $k \geq |l_a-l_b|$, every power of r is a non-negative integer, so no term is singular, and for s orbitals alone the equations reduce to those of paper I.

The point of these variables is that both equations have the same form,

$$ru'' + 2(L+1)u' = g, \tag{5}$$

with $L = l_a$ for an orbital and $L = k$ for a potential, so that one Taylor recurrence (Sec. III A) integrates every unknown. The origin is a regular singular point of Eq. (5), at which the regular solution is fixed by $u(0)$ alone and has slope $u'(0) = g(0)/(2L+2)$. For an orbital this slope is $f_a'(0) = -Zf_a(0)/(l_a+1)$, which is Kato's cusp condition [13], so the cusp is built in exactly.

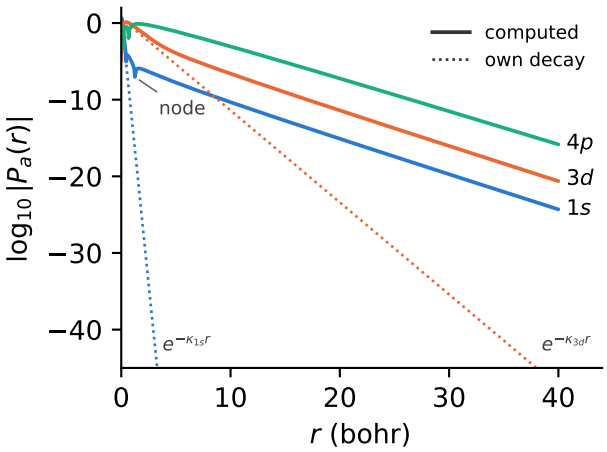


FIG. 1. Inner orbitals of krypton take on the tail of the outermost one. Solid: computed radial functions $|P_a(r)|$ of the $1s$, $3d$ and $4p$ (run A of Table I). Dotted: the decay $e^{-\kappa_a r}$, $\kappa_a = \sqrt{-2\varepsilon_a}$, that the $1s$ and $3d$ would have on their own, drawn from each maximum. Instead both fall off in parallel with the $4p$, because exchange multipoles with $k \geq 1$ couple them to its tail (Sec. II B); at $r = 10$ bohr the $1s$ lies 10^{129} above its own decay line. The marked dip is one of the two exchange-induced nodes of the $1s$ (Table III).

B. One common wall

In paper I each orbital had its own wall, placed where that orbital had decayed, but this fails once p or d shells are present because exchange has a long range. Far outside the electron density $Y^k_{ab}(r) \to m^k_{ab}/r^k$, where $m^k_{ab} = \int_0^\infty s^k P_a P_b\, ds$ is the multipole moment of the pair, so the exchange term in the equation for orbital a behaves as $m^k_{ab} r^{-k-1} P_b(r)$, and what happens next depends on k.

- $k = 0$. For $a \neq b$ the moment is the overlap of the two orbitals, which vanishes. Exchange then has no long-range part, and each orbital decays at its own rate $e^{-\kappa_a r}$, with $\kappa_a = \sqrt{-2\varepsilon_a}$, as in paper I.

- $k \geq 1$. The moment is generally nonzero, so the tail of P_b drives the equation for a, and if b is the outermost orbital its tail decays more slowly than a's own and dominates. Through chains of such couplings every orbital inherits the decay of the highest occupied orbital [10–12].

Figure 1 shows this for krypton: beyond a few bohr every orbital, the $1s$ included, falls off in parallel with the $4p$, so that at $r = 10$ bohr the $1s$ is 10^{-11} of its maximum rather than the 10^{-140} its own decay would give. A wall placed at the $1s$ orbital's own decay length would therefore cut off a tail that is really there. In neon, moving the $1s$ wall changed the residuals of the other equations at the 10^{-6} level, which rules out separate walls at high accuracy.

We therefore use one wall R for all orbitals, with the conditions

$$f_a(R) = 0, \qquad R y^0_{aa}(R) = 1,$$
$$R y^{k\prime}_{ab}(R) + (2k+1)\, y^k_{ab}(R) = 0. \tag{6}$$

The first is the wall itself, the second is the normalisation, since $Y^0_{aa}(R) = \int_0^R P_a^2\, dr$, and the third holds exactly outside the density, where $y^k_{ab} \propto r^{-(2k+1)}$.

A common wall brings two further benefits. The orbitals of one symmetry become eigenfunctions of a single self-adjoint Fock operator on $[0, R]$, so they are exactly orthogonal. The walled problem is also the exact HF problem of an atom in a spherical box, whose energy lies above E_{HF} by an amount that decays as $e^{-2\kappa R}$, with κ that of the highest occupied orbital (paper I), and for D digits we therefore take $R = (D+8)\ln 10/(2\kappa)$.

C. Energy

The energy is needed twice, by the solver, which reports it, and by the certificate, which evaluates it exactly for polynomial orbitals (Sec. IV). Because polynomial orbitals are only nearly orthogonal within a symmetry, we use the closed-shell determinant energy, which does not assume orthogonality:

$$\begin{aligned} E = {} & 2\sum_l (2l+1) \sum_{i,j\in l} D_{ij} h_{ij} \\ & + \sum_{l,l'} (2l+1)(2l'+1) \sum_{\substack{i,j\in l \\ m,o\in l'}} D_{ij} D_{mo} \\ & \times \Big[2R^0(ij,mo) - \sum_k \begin{pmatrix} l & k & l' \\ 0 & 0 & 0 \end{pmatrix}^2 R^k(im,jo) \Big]. \end{aligned} \tag{7}$$

Here $D = S^{-1}$ is the inverse overlap matrix within each block of equal l, $h_{ij} = T_{ij} + V_{ij}$, and

$$R^k(ij,mo) = \iint P_i P_j(r) P_m P_o(s)\, \frac{r^k_<}{r^{k+1}_>}\, dr\, ds \tag{8}$$

are the Slater integrals. Like the s-shell energy of paper I, Eq. (7) depends only on the space spanned by the orbitals of each symmetry. In terms of f, and after the centrifugal term is integrated by parts, the one-electron integrals are

$$T_{ij} = \frac{1}{2}\int_0^R r^{2l+2} f_i' f_j'\, dr, \quad V_{ij} = -Z\int_0^R r^{2l+1} f_i f_j\, dr, \tag{9}$$

because the terms in $r^{2l} f_i f_j$ cancel, $-(l+1)(2l+1) + (l+1)^2 + l(l+1) = 0$, and the boundary terms vanish. The solver evaluates all of these integrals as extra series during the integration.

III. NUMERICAL METHOD

The equations of Sec. II form a nonlinear eigenvalue problem in which n orbitals, their potentials and n orbital energies are all coupled. We solve it in four layers. Taylor series integrate the equations step by step (Sec. III A), and multiple shooting cuts $[0, R]$ into segments so that the growing modes of the inner orbitals never swamp the working precision (Sec. III B). Newton's method then solves the resulting algebraic equations (Sec. III C), from a starting point supplied by a homotopy and a sequence of growing walls (Sec. III D). Appendix A lists every parameter.

A. Taylor recurrences

Substituting $u = \sum_k u_k (r - r_0)^k$ into Eq. (5) gives, about $r_0 > 0$,

$$u_{k+2} = \frac{g_k - (k+1)(k+2L+2)\, u_{k+1}}{r_0 (k+1)(k+2)}, \tag{10}$$

and about $r_0 = 0$, $u_{k+1} = g_k/[(k+1)(k+2L+2)]$. The right-hand sides are products of orbitals and potentials, so their coefficients g_k are Cauchy products and a series of order p costs $O(p^2)$ multiplications per product; krypton has 36 pair products $f_a f_b$ and 73 exchange products $y^k_{ab} f_b$, about 10^2 in all. Each step covers a fixed fraction c of the radius of convergence, which the root test estimates from the last coefficients of the orbital series. The truncation error of a step is then about c^p, and the order p is chosen to put it below the target accuracy.

One subtlety matters for heavy atoms. The root test sees only the current solution, and because far out every orbital follows the slow common tail, near the wall it would allow a single step across a whole segment. Newton's method, however, also works with each orbital's own fast modes $e^{\pm\kappa_a r}$, which make up the Jacobian columns and the corrections, and at 100 digits such a step resolves them to only 15–20 digits, which is enough to make Newton's method fail. At order p the series of $e^{\kappa r}$ behaves as if its radius of convergence were $p/(e\kappa)$, so we also cap every step at $c\,p/(e\kappa_1)$, with κ_1 the largest κ_a; the cap leaves the energies of the light atoms unchanged to every digit.

B. Multiple shooting

Over $[0, R]$ the growing mode of the krypton $1s$ rises by $e^{\kappa_{1s} R} = 10^{1701}$, so shooting from the origin would need that many extra digits. Multiple shooting [14] avoids this by cutting $[0, R]$ into segments, $0 = r_0 < r_1 < \cdots < r_J = R$, and integrating each one independently from its own starting state. The unknowns are then the start values $f_a(0)$ and $y^k_{ab}(0)$, the full state $(f_a, f_a', y^k_{ab}, y^{k\prime}_{ab})$ at each interior knot and the orbital energies ε_a, while the

equations require the state to be continuous at every interior knot and the conditions (6) to hold at R. With n orbitals and n_y potentials there are $n_0 = n + n_y$ start values, $m = 2n_0$ unknowns per knot and n energies; for krypton $n_y = 47$, $m = 110$ and $J = 67$, so the system has 7323 unknowns.

The segments are kept short enough that no growing mode rises, relative to the bulk density $e^{-\kappa r}$, by more than a factor 10^g across one. An error in a segment's starting state is then amplified by at most 10^g, and the linear solve loses at most another g digits, so for d target digits we work with $d + 2g + 40$ digits. This margin depends on g but not on how stiff the atom is, because a stiffer atom simply needs more segments, and inner orbitals shorten the segments only where they are themselves large (Appendix A).

C. Newton's method

We solve the system by a chord Newton iteration [15], which keeps one Jacobian and refreshes it only when a step fails to shrink the correction threefold. The Jacobian is formed by forward differences with all columns in one parallel batch, in which a column for a knot state needs only one segment to be integrated again and a column for an ε_a needs all of them. Its structure is block bidiagonal, with $-I$ coupling each segment to the next knot, plus dense columns for the ε_a, so sparse elimination with row equilibration and partial pivoting keeps the fill inside the band.

Because the chord iteration converges as long as the Jacobian is accurate enough to contract, during the wall continuation (Sec. III D) we build and factor it at about $4g + 30$ digits, with a matching lower Taylor order. The residuals are always evaluated at full precision, so the converged solution is unchanged, and a cheaper Jacobian that fails is rebuilt at full precision.

D. Starting point and continuation

Newton's method needs a starting point close to the ground state, and the equations for a heavy atom have many other solutions, including ones in which the wall pushes an orbital to positive energy. We therefore reach the ground state in two stages.

The first is a homotopy in the electron–electron interaction: every electron–electron term in Eq. (3) is multiplied by t, and the solution is followed from $t = 0$, where the hydrogenic orbitals are exact, to $t = 1$. Each step in t changes the potentials by a finite amount, which a segment amplifies by 10^g, so here we keep $g = 1$ and a modest target of 10 digits. Meanwhile the wall moves with the orbitals, at $R = \max_a(2n_a + 6)/\kappa_a(t)$, where every hydrogenic $r^{n-1}e^{-\kappa r}$ has fallen to about 10^{-3} of its peak, so that no orbital is squeezed.

The second stage moves the wall out to its final value, with each step targeting the digits its wall can deliver, $\min(D, 2\kappa R/\ln 10) + 10$. Moving the wall changes the solution only by about $e^{-\kappa R}$ near the old wall, so these steps can afford long segments, with g up to 30. The starting guess on the new knots is the previous solution integrated out to them, continued beyond the old wall with each orbital's own logarithmic derivative and each potential's multipole tail.

E. Implementation

The orchestration (homotopy, continuation, Newton's method and knot placement) is a few hundred lines of Python with `gmpy2`, while the numerical work, the integration of a segment and the sparse LU, is done by a small C library on MPFR [16] and GMP [17]. In the convolutions each product of two MPFR numbers is formed exactly and added to a fixed-point accumulator, and the sum is rounded once [18]; this is more accurate than a multiply-and-add loop, which rounds $2k + 1$ times, and 1.8–2.4 times faster at 170–400 bits. Segments and Jacobian columns are spread over a pool of processes, one per core, and a test checks the C integration against a reference Python implementation of the same recurrences.

IV. CERTIFICATE

The certificate is a separate program that reads the orbitals written by the solver and proves an upper bound on E_{HF} in exact arithmetic. It rests only on the variational principle, by which the energy of any determinant of admissible orbitals lies above E_{HF}, and so it needs two things: admissible orbitals that are known exactly, and every integral of Eq. (7) evaluated exactly.

The first is straightforward. The solver writes each f_a as a piecewise polynomial whose knots and coefficients are dyadic rationals, and the certificate checks exactly that the pieces tile $[0, R]$ and imposes continuity and $f_a(R) = 0$ exactly, which makes the orbitals admissible.

For the second, every integrand is a polynomial on each piece, so the overlaps and the one-electron integrals (9) are exact, and the Slater integrals need only one more step. For the pair densities $q = (ij)$ and $p = (mo)$ we write $R^k(q, p) = C^k(q, p) + C^k(p, q)$, with

$$C^k(q,p) = \int_0^R r^{l_q+1-k} f_i f_j(r)\, Q_p^k(r)\, dr,$$
$$Q_p^k(r) = \int_0^r s^{l_p+k+2} f_m f_o\, ds, \tag{11}$$

and $l_q = l_i + l_j$. Here the multipole factor r^{-k-1} has combined with $P_i P_j = r^{l_q+2} f_i f_j$ into r^{l_q+1-k}, and since the triangle rule gives $k \le l_q$, this power is non-negative and the integrand is again a polynomial. This is why we carry $f = P/r^{l+1}$ rather than P: with P, these integrands

would contain negative powers of r, whose integrals are logarithms. The inner integral Q_p^k is accumulated across the pieces as an exact antiderivative.

Finally, $D = S^{-1}$ is formed exactly in each symmetry block and the energy is rounded towards $+\infty$ only when it is printed, so the printed number is a rigorous upper bound on E_{HF}. For s shells alone the program returns the same rational as the certificate of paper I. Since the pieces are independent apart from the running integrals Q_p^k, they are processed in parallel, each returning its own contributions, and a final prefix sum over the pieces supplies the running integrals; the parallel program returns the identical rational.

V. RESULTS

A. Energies

Table I lists the production runs and Table II the results. As in paper I, run B has a larger wall, so its own wall error is about 10^{-10} of run A's, and its truncation and rounding are set at least 10 digits further down. The difference between run A's certified bound and run B is therefore the distance of the bound from E_{HF}, and the number in the "bound" column is the number of correct significant figures of the certified value. For neon the certificate of the 500-digit run gives

$$\begin{aligned} E_{\mathrm{HF}}(\mathrm{Ne}) \le -128.547\,098\,109\,382\,042\,376\,475 \\ 781\,341\,993\,899\,003\,173\,076 \\ 242\,596\,929\,050\,687\ldots\ E_{\mathrm{h}}, \end{aligned} \tag{12}$$

correct to 504 significant figures; Appendix B has them all.

Three checks are independent of the A/B comparison.

- **Published values.** All 25 digits of Cinal's quadruple-precision energies [8] agree with ours after rounding, for Ne, Mg, Ar, Ca, Zn, Kr, Sr, Pd, Cd, Xe, and so do his 22–25-digit energies of the highest occupied orbital.
- **Orthogonality.** The canonical orbitals of each symmetry are orthogonal to the working precision, for example $|S_{ij}| \le 8 \times 10^{-114}$ for krypton, as the common wall requires. The determinant energy (7) does not assume this.
- **Upper bounds.** Every certified bound lies above the energy of the more accurate run B, as it must.

B. Exchange-induced nodes

The induced tails of Sec. II B have a further consequence. Where an inner orbital's own part has decayed to the size of its induced tail, the two can cancel, and the orbital acquires a node that a hydrogenic orbital does not have. Such nodes have been reported for the $1s$ orbitals of argon, zinc and krypton [8, 19] and in the $2s$ of krypton [8, 21], and traced to exchange [20]. In basis-set calculations they are hard to distinguish from the spurious nodes of a finite basis. Here they come out of the solution with no special treatment (Table III). Magnesium, whose orbitals are all s orbitals, has none, as the $k = 0$ argument of Sec. II B requires. Argon has one, in the $1s$; calcium and zinc have them in the $1s$ and $2p$, krypton and palladium also in the $2s$, strontium and cadmium also in the $3p$ and $3d$, and xenon also in the $3s$. The krypton $1s$ changes sign at $r = 0.45, 1.26$ bohr, where it is 5–7 orders of magnitude below its maximum. Because the error of a FROST orbital is the wall's, and that is exponentially small in the core, these node positions agree between runs A and B to about as many figures as the energies.

C. Computational cost

The calculations ran one at a time on the 12 cores of an Apple M4 Pro laptop. The cost of a run is dominated by the Jacobians, each of which costs $m + n$ integrations of the full range in lower precision, while a Newton iteration costs one. At 100 digits run A took 7 min for calcium, 37 min for krypton and 3.9 h for xenon, and the certificates of the atoms from calcium to xenon took 0.2–2.1 h. The 500-digit neon pair and its certificate took 26, 41 and 17 min.

VI. DISCUSSION

With one wall, multiple shooting and a homotopy start, FROST computes the HF limit of any closed-shell atom to essentially any number of digits, in time polynomial in the number of digits, and certifies it. The difficulty that heavy atoms add is stiffness, not dimension. The inner shells' growing modes are confined to segments, and the extra precision this costs depends on the segment growth 10^g, not on the ratio of the inner and outer decay rates. The common wall, needed because of the induced tails, also gives orthogonal canonical orbitals, which removes the conditioning problems that the separate walls of paper I caused for beryllium.

The certificate needed only the observation that $f = P/r^{l+1}$ keeps every integrand polynomial. It is independent of the solver and indifferent to how the orbitals were obtained, and every value in Table II is a rigorous upper bound. As in paper I the bound is one-sided, and the distance to E_{HF} rests on the independent run B and the exponential decay of the wall error.

The remaining closed-shell atoms of Cinal's set, Ba, Yb, Hg, Rn, Ra and No, add an f shell ($l = 3$) or more electrons. Neither changes the formulation: the multipoles go up to $k = 6$, and the certificate is unchanged. The cost per digit grows with the number of potentials,

TABLE I. Production runs. Run A targets D digits with $c = 0.15$ and is certified; run B is an independent run with $D + 10$ digits (a larger wall), $c = 0.25$ and 30 instead of 40 guard digits, so with a different mesh, order, knots and precision. R: wall of run A (bohr); J: segments; p: Taylor order; bits: MPFR working precision. "A–B" is the number of significant figures to which the energies of runs A and B agree, and "bound" that to which the certified bound from run A agrees with run B; in every case the bound lies above run B. "prev.": the best previous number of significant figures.

Z	atom	configuration	D	R	J	p	bits	A–B	bound	prev.
10	Ne	$[He]2s^2 2p^6$	500	448.47	54	626	2026	504.6	504.6	30 [9]
12	Mg	$[Ne]3s^2$	100	174.78	31	140	647	103.7	103.7	30 [9]
18	Ar	$[Ne]3s^2 3p^6$	100	114.38	31	140	648	105.4	105.4	30 [9]
20	Ca	$[Ar]4s^2$	100	198.84	59	140	647	103.5	103.5	25 [8]
30	Zn	$[Ar]3d^{10}4s^2$	100	162.58	71	140	653	105.0	105.0	25 [8]
36	Kr	$[Ar]3d^{10}4s^2 4p^6$	100	121.44	67	140	647	105.8	105.8	25 [8]
38	Sr	$[Kr]5s^2$	100	208.14	112	140	659	103.9	103.9	25 [8]
46	Pd	$[Kr]4d^{10}$	100	151.69	99	140	660	106.5	106.6	25 [8]
48	Cd	$[Kr]4d^{10}5s^2$	100	170.84	125	140	647	105.2	105.2	25 [8]
54	Xe	$[Kr]4d^{10}5s^2 5p^6$	100	130.03	103	140	653	105.9	105.9	25 [8]

TABLE II. Certified HF energies (hartree), truncated (each value is an upper bound on E_{HF}), and energies of the highest occupied orbital, truncated to 25 significant figures. Appendix B gives every correct figure of the bounds.

atom	$E_{\mathrm{HF}} \leq$	$\varepsilon_{\mathrm{HOMO}}$
Ne	−128.547 098 109 382 042 376 475 781 341 993 899 003 173 076 . . .	−0.850 409 650 347 267 227 432 801 7 . . .
Mg	−199.614 636 424 506 710 474 741 636 103 218 994 331 839 478 . . .	−0.253 052 582 006 473 108 897 913 3 . . .
Ar	−526.817 512 802 723 355 018 766 093 533 797 981 273 149 792 . . .	−0.591 017 409 350 350 745 054 566 5 . . .
Ca	−676.758 185 924 513 454 948 055 751 766 316 428 937 000 333 . . .	−0.195 529 692 467 776 181 004 838 2 . . .
Zn	−1777.848 116 191 361 069 778 442 503 292 099 775 989 467 52 . . .	−0.292 507 146 402 791 438 403 119 9 . . .
Kr	−2752.054 977 345 576 454 394 449 138 238 326 238 385 587 02 . . .	−0.524 186 681 026 595 659 788 005 1 . . .
Sr	−3131.545 686 438 557 184 434 124 153 949 571 921 023 530 95 . . .	−0.178 456 446 851 434 750 663 585 3 . . .
Pd	−4937.921 024 070 101 572 932 289 803 807 301 519 419 766 54 . . .	−0.336 002 013 665 775 551 570 730 8 . . .
Cd	−5465.133 142 530 086 433 672 818 631 277 862 147 841 082 70 . . .	−0.264 855 912 553 887 068 303 327 0 . . .
Xe	−7232.138 363 871 840 888 209 289 216 559 060 407 211 433 76 . . .	−0.457 290 057 851 833 026 468 458 2 . . .

roughly as the square of the number of orbitals, and with the stiffness through the number of segments. Open shells need the restricted open-shell energy expression with its additional couplings. For diatomic molecules the radial ODEs become partial differential equations in prolate spheroidal coordinates, and the method would need a two-dimensional analogue of the Taylor segments.

DATA AND CODE AVAILABILITY

The solver (`oracle_ns.py` with the C kernel `taylor.c`), the certificate (`certify.py`), a test, the run summaries and the script generating every table and figure of this paper are in Ref. [22], together with the certificate output and logs of every run in Table I and the SHA-256 checksums of their orbital files. The orbitals themselves, runs A and B for every atom (1.4 GB compressed), are archived in Ref. [23], so that any bound in Table II can be checked again with `certify.py` without repeating the run.

Appendix A: Parameters

These choices set the cost and the robustness of a run. Runs A and B use different values (Table I), and their agreement shows that none of them affects the result beyond the stated accuracy.

- **Walls.** Every wall, including the intermediate ones, is rounded up to a multiple of 1/64 bohr, so that knots are exact dyadic rationals.

- **Taylor steps.** A step is $h = c\rho$, rounded down to a multiple of 2^{-16}, with $\rho = \min_{a,k} \left[\max(|f_{a,0}|, |f_{a,1}|)/|f_{a,k}| \right]^{1/k}$ over the last nine coefficients of each orbital series. The scale $\max(|f_{a,0}|, |f_{a,1}|)$ keeps an expansion point at a node from faking a small radius. The order is $p = \lceil (d+5)/\log_{10}(1/c) \rceil$ for d digits, and $h \leq c\,p/(e\kappa_1)$ (Sec. III A). Run A uses $c = 0.15$, run B $c = 0.25$.

- **Segments.** An orbital i limits a segment that

TABLE III. Exchange-induced nodes (bohr) of inner orbitals: sign changes of P_a beyond the $n_a - l_a - 1$ of the hydrogenic orbital, located by bisection to 100 decimal places on the certified piecewise polynomials of run A and truncated. "A–B": significant figures to which runs A and B agree (>99: every figure of the bisection). Each row is one node: an orbital that changes sign more than once has a row for each node, from the inside out, so the three rows under xenon 1s are three nodes of its single 1s orbital. Atoms continue in the right-hand panel. Cinal's values [8], given to ten figures for Ar, Zn and Kr, agree with every figure after rounding.

atom	orbital	node	A–B	atom	orbital	node	A–B
Ar	1s	1.093 265 789 408 326 436 857 636 973 32 . . .	>99	Cd	1s	0.325 419 597 872 657 101 813 841 819 584 . . .	>99
Ca	1s	0.907 998 824 349 994 249 762 216 842 103 . . .	>99			0.722 048 860 293 642 719 523 419 736 970 . . .	>99
	2p	3.308 277 004 757 110 108 545 902 600 08 . . .	>99		2s	0.876 322 822 413 437 451 065 913 628 413 . . .	>99
Zn	1s	0.558 872 044 838 827 557 936 068 541 392 . . .	>99		2p	0.904 532 872 588 058 835 214 850 283 131 . . .	>99
	2p	5.297 047 587 673 908 948 294 641 013 22 . . .	>99			5.585 301 466 066 447 304 688 329 136 54 . . .	>99
Kr	1s	0.448 341 935 663 648 833 694 790 412 998 . . .	>99		3p	4.314 942 357 534 312 714 731 960 672 36 . . .	>99
		1.255 565 714 380 793 982 191 685 131 25 . . .	>99		3d	4.740 476 277 615 026 747 379 116 848 96 . . .	>99
	2s	1.506 285 094 245 466 491 324 205 335 15 . . .	>99	Xe	1s	0.286 842 855 951 185 928 823 876 868 287 . . .	>99
	2p	2.238 770 728 246 546 684 612 198 555 81 . . .	>99			0.586 127 622 355 550 164 135 872 896 792 . . .	>99
Sr	1s	0.420 452 597 105 506 063 859 532 146 599 . . .	>99			1.588 543 061 635 355 396 989 725 406 38 . . .	>99
		1.085 857 283 644 387 382 160 310 745 77 . . .	>99		2s	0.736 294 303 620 627 506 677 792 367 441 . . .	>99
	2s	1.318 267 473 098 623 561 755 925 107 39 . . .	>99			1.702 541 247 816 521 255 286 768 175 37 . . .	>99
	2p	1.923 492 641 655 152 973 985 345 582 53 . . .	>99		2p	0.745 742 720 822 419 294 191 289 945 686 . . .	>99
		3.195 279 907 173 433 981 256 852 267 81 . . .	98			2.958 324 503 852 447 794 381 812 253 21 . . .	>99
	3p	3.752 447 606 549 993 764 366 032 907 47 . . .	98		3s	1.962 961 123 101 139 570 202 164 612 02 . . .	>99
	3d	6.610 669 617 583 112 978 116 189 708 39 . . .	98		3p	2.423 985 240 309 643 641 646 187 050 88 . . .	>99
Pd	1s	0.340 601 465 925 598 401 718 174 746 614 . . .	>99		3d	2.103 622 207 445 040 314 154 307 100 18 . . .	>99
		0.774 335 880 052 257 872 212 211 367 782 . . .	>99				
		10.194 294 095 354 715 726 538 808 753 8 . . .	>99				
	2s	0.936 362 744 315 489 794 629 259 353 134 . . .	>99				
	2p	0.985 254 722 908 618 840 284 238 969 629 . . .	>99				

starts at r to

$$L_i = \frac{g \ln 10 + \min\big[(\kappa_i - \kappa)\max(r - n_i/\kappa_i, 0),\ 4 \ln 10\big]}{\kappa_i + \kappa},$$

and the segment is $\min_i L_i$. Beyond its own radius n_i/κ_i orbital i has fallen below the bulk density. It may grow back by as much as it has fallen, but by at most 10^4, the level of its induced tail.

- **Precision.** $P = d + 2g + G$ digits, with guard $G = 40$ for run A and 30 for run B.

- **Jacobian.** Forward differences with relative step $10^{-P_J/2}$, so that about $P_J/2$ digits are correct, and Taylor order $\min(p, \lceil (P_J/2 + 5)/\log_{10}(1/c) \rceil)$. In the homotopy $P_J = P$; in the wall continuation $P_J = \min(P, 4g + 30)$. If this cheaper Jacobian is singular, gives a step whose residual is not finite, or stops the chord iteration from contracting, it is rebuilt with $P_J = P$ at full order. Later stages, whose walls are larger, then start with $P_J = P$.

- **Homotopy.** $g = 1$ and a 10-digit target. The step Δt starts at 1/32, grows by a factor 1.5 (up to 0.25) after a step that needed one Jacobian, and halves after a failure. The start values and the ε_a are extrapolated linearly in t. The wall $\max_a(2n_a + 6)/\kappa_a(t)$ never decreases. A wall at $8/\kappa$, adequate up to $n = 4$, squeezes a hydrogenic 5s orbital to positive energy; any step that ends with an $\varepsilon_a \geq 0$ is rejected.

- **Wall continuation.** The wall grows by a factor ϱ per stage. ϱ starts at 1.2, grows by 1.25 (up to 2) after a stage that needed one Jacobian, and returns to 1.2 after a harder stage; a stage that fails is repeated with $\varrho = 1.2$. Once the final wall is within 1.1ϱ, the stage goes straight to it. A stage from $R_{\rm old}$ to R uses $g = \min[30, \max(1, \kappa \max(R_{\rm old}, R/1.2)/(2 \ln 10))]$. The guess is continued from $\max(R_{\rm old} - 1/\kappa, R_{\rm old}/2)$ with logarithmic derivatives no larger than $-\kappa/2$.

- **Accumulator.** Its last bit lies 128 bits below the largest product. Sums of 4096 terms or more, or operands beyond 8192 bits, fall back to an MPFR multiply-and-add loop.

Appendix B: Digits

Certified upper bounds from runs A of Table I, truncated to their correct significant figures (truncation keeps every value an upper bound).

Ne (504 s.f.): $E_{\mathrm{HF}} \leq -128.547\,098\,109\,382\,042\,376\,475\,781\,341\,993\,899\,003\,173\,076\,242\,596\,929\,050\,687\,475\,797\,069\,115\,074\,707\,767\,260\,398\,406\,505\,305\,247\,420\,940\,746\,468\,946\,609\,084\,287\,991\,634\,015\,929\,780\,188\,747\,771\,659\,181\,986\,019\,093\,586\,199\,911\,964\,925\,483\,280\,759\,270\,848\,950\,192\,784\,003\,812\,355\,908\,269\,129\,125\,119\,142\,646\,533\,535\,399\,791\,205\,303\,241\,987\,858\,693\,316\,174\,748\,568\,889\,976\,789\,100\,017\,394\,251\,843\,704\,767\,129\,492\,712\,056\,128\,576\,187\,952\,639\,827\,247\,380\,409\,309\,374\,136\,477\,816\,843\,986\,495\,272\,413\,166\,974\,310\,452\,350\,947\,426\,230\,945\,816\,059\,558\,769\,805\,229\,509\,598\,079\,161\,533\,840\,788\,937\,616\,063\,486\,519\,041\,893\,812\,573\,813\,831\,336\,475\,429\,935\,609\,599\,657\,159\,733\,402\,190$

Mg (103 s.f.): $E_{\mathrm{HF}} \leq -199.614\,636\,424\,506\,710\,474\,741\,636\,103\,218\,994\,331\,839\,478\,168\,286\,337\,445\,498\,301\,924\,473\,587\,677\,543\,045\,017\,664\,932\,235\,196\,369\,459\,6$

Ar (105 s.f.): $E_{\mathrm{HF}} \leq -526.817\,512\,802\,723\,355\,018\,766\,093\,533\,797\,981\,273\,149\,792\,066\,208\,360\,226\,091\,841\,366\,064\,951\,149\,381\,765\,608\,575\,025\,568\,098\,087\,838\,734$

Ca (103 s.f.): $E_{\mathrm{HF}} \leq -676.758\,185\,924\,513\,454\,948\,055\,751\,766\,316\,428\,937\,000\,333\,163\,970\,672\,672\,400\,483\,219\,679\,908\,510\,662\,795\,808\,217\,091\,633\,190\,749\,607\,1$

Zn (104 s.f.): $E_{\mathrm{HF}} \leq -1777.848\,116\,191\,361\,069\,778\,442\,503\,292\,099\,775\,989\,467\,529\,793\,653\,248\,391\,927\,884\,196\,729\,996\,275\,339\,929\,455\,468\,065\,341\,340\,623\,198\,0$

Kr (105 s.f.): $E_{\mathrm{HF}} \leq -2752.054\,977\,345\,576\,454\,394\,449\,138\,238\,326\,238\,385\,587\,027\,503\,759\,496\,356\,329\,506\,682\,845\,233\,924\,066\,208\,567\,490\,469\,066\,223\,590\,318\,04$

Sr (103 s.f.): $E_{\mathrm{HF}} \leq -3131.545\,686\,438\,557\,184\,434\,124\,153\,949\,571\,921\,023\,530\,956\,865\,822\,141\,483\,649\,630\,053\,101\,650\,882\,063\,430\,634\,824\,075\,045\,921\,404\,228$

Pd (106 s.f.): $E_{\mathrm{HF}} \leq -4937.921\,024\,070\,101\,572\,932\,289\,803\,807\,301\,519\,419\,766\,542\,505\,417\,631\,137\,311\,923\,258\,591\,744\,554\,570\,785\,963\,821\,360\,229\,625\,624\,576\,188$

Cd (105 s.f.): $E_{\mathrm{HF}} \leq -5465.133\,142\,530\,086\,433\,672\,818\,631\,277\,862\,147\,841\,082\,707\,966\,792\,559\,616\,118\,585\,695\,723\,792\,132\,872\,583\,758\,812\,408\,581\,757\,610\,273\,73$

Xe (105 s.f.): $E_{\mathrm{HF}} \leq -7232.138\,363\,871\,840\,888\,209\,289\,216\,559\,060\,407\,211\,433\,760\,515\,718\,488\,246\,631\,050\,704\,195\,202\,168\,260\,446\,688\,360\,588\,265\,374\,199\,373\,01$

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